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SkyGuard: A Hierarchical SLM–LLM Threat Hunting Framework for the OpenC3+NASA cFS Space-Software Stack

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***Abstract*—Prior LLM-based security frameworks for Space-Air-Ground Integrated Networks (SAGIN) follow two patterns: multi-agent peer-LLM systems and edge-ML / central-LLM hybrids. A recent intrusion-detection survey identifies a third pattern as a future direction—a *capacity-graded hierarchy* pairing generative, tool-calling Small Language Models (SLMs) at the edge with a larger central LLM orchestrator—but it has not been realized. We argue that this pattern is not stylistically preferable but architecturally necessary in SAGIN: short LEO contact windows, bandwidth-rationed uplinks, and the published verdict that ground-based IDS is ineffective against real-time intra-satellite threats together rule out the central-analyzer pattern. We instantiate the pattern as SkyGuard, specialized to the OpenC3 ground-segment platform and NASA Core Flight System (cFS) space-segment software. Per-stack edge SLMs investigate locally via snapshot integrity verification, bounded-window log queries, and semantic cross-dimension reasoning; a central LLM correlates MITRE ATT&CK-tagged summaries into APT chains and dispatches directives carrying abstract behavioral patterns only—never concrete indicators. Under leave-one-out cross-validation (LOO-CV) over 30 proof-of-concept (PoC) attacks (six previously unreported deep-cFS exploits) and five APT chains, SkyGuard is the only deployable configuration evaluated that detects multi-stage chains at all: a static-scripts baseline and an edge-SLM-only baseline both score zero on every chain, while a privileged full-information central LLM achieves higher absolute recall under a link premise SAGIN cannot satisfy. SkyGuard reaches 0.46 macro category recall (0.65 excluding LOO orphan-application categories, where the held-out fold leaves zero same-application training examples by construction) and 0.43 macro chain recall under strict LOO scoring.**

***Index Terms*—SAGIN, threat hunting, large language models, small language models, edge intelligence, satellite security**

I. Introduction

Space-Air-Ground Integrated Networks (SAGIN) fuse low-Earth-orbit (LEO) and geostationary satellites, UAV / High-Altitude Platform Station (HAPS) relays, and terrestrial infrastructure into a single heterogeneous fabric [1], [2]. The same architectural openness that enables global coverage also expands the attack surface: an adversary who compromises a satellite ground station may pivot laterally and ultimately affect space operations, and recent surveys highlight unresolved security, privacy, and trust-management challenges in SAGIN deployments [3], [4].

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Why a hierarchy, not a centralized analyzer. The defining operational constraint in SAGIN security is not privacy preference but link feasibility. A single LEO ground station typically has 5–15 minute contact windows per pass with four to six passes per day for a given satellite-station pair [5], and the satellite-to-ground link is bandwidth-rationed—uplink particularly so [6], [7]. Xiao et al. in *IEEE Journal on Selected Areas in Communications* state the implication directly: ground-based intrusion detection is *"fundamentally constrained by the high-latency and bandwidth-limited satellite-to-ground link, rendering [it] ineffective against real-time, intra-satellite threats"* [8], and they accordingly advocate decoupling security analysis into ground-based statistical estimation and on-board real-time execution [8]. The same conclusion is reached independently from the orbital edge computing literature [9], [10], which documents that downlinking raw observations for ground processing is throughput- and contact-window-bounded and motivates in-orbit inference for time-sensitive tasks. Beyond infeasibility, streaming raw operational telemetry across the link *also* expands the attack surface: telemetry, tracking, and control (TT&C) data in transit are the exact category of artifact that adversaries target [8], so a security architecture that depends on bulk-egressing this data to a central analyzer inverts the threat model it claims to defend. These constraints rule out the central-analyzer pattern that works for terrestrial SOCs and motivate a hierarchy in which heavy reasoning lives as close as possible to where operational data is generated, with only compact, sanitized summaries traversing the link.

Two complementary lines of work motivate this paper. First, LLM-based security for SAGIN has so far taken the form of *multi-agent LLM systems*: Qin et al. [11] propose LLM-6GNG with chain-of-thought reasoning across dynamic multi-agent LLMs, and Cao et al. [12] extend this to zero-trust SAGIN with LLM-SA, a multi-agent situation-awareness method. In both, every agent is itself an LLM and the architecture is horizontal: peer agents collaborate, rather than a smaller model handling local detail under a larger correlator. Second, beyond SAGIN, a separate line of work pairs *classical lightweight ML or compressed encoders at the edge* with *a central LLM analyzer*. *Distributed Threat Intelligence at the Edge Devices* [13] routes only suspicious traffic and log-change summaries from edge

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ML models to a central LLM; Karunanayake et al. [14] formalize LLM-driven adaptive policy orchestration for host-based IDS; ZeroDay-LLM [15] pairs compressed BERT encoders at the edge with a centralized transformer reasoning engine. In all three the edge tier is *not* a generative language model and cannot, by itself, run an investigation trajectory.

A recent survey of LLMs for network intrusion detection by Yang et al. [16] explicitly identifies a third pattern as a future direction: deploying *Small Language Models* at the edge as first-class agents and pairing them with a larger central LLM, so that both tiers are generative language models distinguished by capacity and role. To our knowledge no published security framework has yet realized this capacity-graded SLM/LLM hierarchy, and no work has applied any form of LLM-based or agentic threat hunting to the open-source software stack that dominates real ground- and space-segment operations—the OpenC3 command-and-control platform [17], [18] and NASA cFS [19]. Public security analyses of these stacks have cataloged multi-layer vulnerabilities spanning protocol injection, credential abuse, and persistent backdoors [20], but no end-to-end automated detector has been demonstrated. SkyGuard addresses both gaps.

Our contributions are of two kinds: an *architectural-necessity argument* that reframes an existing pattern for the SAGIN setting, and two *concrete design mechanisms* plus a reproducible artifact contribution that, to our knowledge, do not appear in prior work. We do *not* claim novelty for hierarchical LLM organization in general, retrieval-augmented generation, tool-calling agents, or LLM-based Sigma-rule generation—each exists in the literature we cite; the specific deltas are stated below and revisited in Section VII.

1) **Architectural-necessity argument.** Using published LEO contact-window and uplink-bandwidth constraints [5]–[7] and the STARDIS verdict that ground-based IDS is ineffective against real-time intra-satellite threats [8], we argue that the capacity-graded SLM/LLM pattern identified as a future direction by Yang et al. [16] is not stylistically preferable but architecturally required in SAGIN: alternative patterns (Section II) cannot satisfy the link budget. We instantiate the pattern accordingly as SkyGuard (Section IV).
2) **Locality-preserving directive design** (Section V, *new*). Downward `HUNT` directives from the central LLM carry only abstract behavioral patterns drawn from ATT&CK/SPARTA [21], [22]; receiving SLMs discover all concrete indicators of compromise (IOCs) locally through bounded tool calls, and the central LLM never holds a concrete IOC. This narrows the inter-tier channel to fit within contact-window throughput (quantified in Section VI) and keeps raw telemetry off the in-transit attack surface identified in [8]. Yang et al. specify neither the summary schema nor the directive semantics; both are original to this work.
3) **Two-tier Sigma harvest workflow** (Section IV, *new*). Rule generation is split by redaction state: *Tier A* pre-redaction (operationally precise, retained locally, SIEM-deployable) and *Tier B* post-redaction (abstract, portable, shareable with Information Sharing and Analysis Centers (ISACs) and the ATT&CK community). Prior LLM-driven Sigma generation [29] produces a single rule tier; bifurcation by redaction state is, to our knowledge, new.
4) **Reproducible artifact for OpenC3+cFS** (Section VI). A 30-PoC catalog on the OpenC3 ground-station and NASA cFS space-segment stacks—the first such catalog we are aware of for these stacks—including six previously unreported deep-cFS exploits (Memory Manager arbitrary R/W, EVS stealth, CDS persistence, ES exec, CS checksum bypass, CFDP dual-channel backdoor) and five APT chains spanning four correlation scopes, reported under LOO-CV to separate memorisation from generalization.

II. Background and Related Work

LLMs for threat hunting. Thelma [23] and its successor iThelma [24] introduced LLM agents that ingest threat-intelligence playbooks, generate hunt scripts in a sandbox, and refine them through execution feedback. DistilGuard [25] applied LLMs to federated-learning poisoning detection, while APTMORPH [26] used generative AI with reinforcement learning to anticipate APT technique evolution. None of these works addresses the distributed, bandwidth-constrained, or mission-critical confidentiality requirements of SAGIN.

Pattern A: multi-agent LLM frameworks. Qin et al. [11] propose LLM-6GNG and 6G-INST, an LLM-based self-evolving security framework for 6G SAGINs in which multiple LLM agents reason via chain-of-thought. Cao et al. [12] extend this to zero-trust SAGIN with LLM-SA, a multi-agent situation-awareness method, releasing the SAG-Attack simulator alongside. In both, the agents are peer LLMs of comparable class; the architecture is horizontal. This pattern is well suited to the strategic-reasoning and threat-information-synthesis tasks addressed by those works, but it does not address the question of where the heavy reasoning should live relative to where the operational data is generated: every agent in a multi-agent LLM system is a full LLM, and there is no inherent privacy or bandwidth boundary between them.

Pattern B: edge classical-ML or compressed-encoder front-end with central LLM. A separate line of work places *non-generative* models at the edge. *Distributed Threat Intelligence at the Edge Devices* [13] runs lightweight ML models on edge devices and forwards only suspicious-event summaries to a central LLM analyzer. Karunanayake et al. [14] formalize LLM-driven adaptive policy orchestration for host-based IDS in IoT environments. ZeroDay-LLM [15] pairs compressed BERT encoders at the edge with a centralized transformer reasoning engine for zero-day IoT/cloud detection. A CNS 2025 study [27] additionally shows that locally deployed sub-7B to 14B models can approach cloud-frontier performance on 5G CVE classification while eliminating data exfiltration risk. These designs deliver the privacy and latency benefits of edge

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processing, but the edge tier cannot itself run an investigation trajectory: a non-generative encoder or classical ML model produces an embedding or anomaly score, not a chain of tool calls reasoning over its own outputs. All agentic behavior remains centralized. On the chain-detection axis a Pattern-B edge tier faces an additional SAGIN-specific limitation: the ATT&CK-tagged summaries needed for upstream correlation cannot be composed by a scoring function, and the link budget rules out forwarding raw signals to the central LLM to have it compose them there. A classical-ML edge classifier therefore inherits the failure mode our edge-SLM-only baseline B2 exhibits on chain detection (Section VI), without B2's ability to recover read-only, transient, or in-memory single-stage attacks through semantic tool-call reasoning; the B1/B2/B4 axis in our evaluation brackets the Pattern-B design space on both metrics.

Pattern C: capacity-graded SLM/LLM hierarchy. Yang et al.'s recent survey of LLMs for network intrusion detection [16] identifies a third pattern as a future direction: deploying generative Small Language Models at the edge as first-class agentic detectors and pairing them with a larger central LLM for correlation. Both tiers are language models; what distinguishes them is parameter count, role, and the data they may see. To our knowledge, no published security framework has yet realized this pattern. SkyGuard does, with two additional specializations to the SAGIN context: (i) a locality-preserving directive regime motivated by the link constraints in Section I, in which the central LLM never holds concrete IOCs (Section V), and (ii) a two-tier Sigma harvest that bifurcates rule generation by redaction state (Section IV).

Satellite/SAGIN software security. Ear et al. [20] characterise cyber attacks against space systems from public records. Sigma rules [28] provide a generic SIEM detection format that LLMCloudHunter and others [29] have begun to auto-generate from open-source threat intelligence. SkyGuard differs by (i) operating on a real OpenC3+NASA cFS testbed with a reproducible 30 PoC catalog rather than mining public incidents, and (ii) bifurcating Sigma generation by redaction state so that operational precision and community portability are harvested simultaneously.

III. Threat Model and PoC Attack Catalogue

Five-layer defensive model. The system spans three network strata: a *space segment* (satellite bus and payload, with NASA cFS as flight software), an *air segment* (UAV/HAPS relays treated here as forwarding nodes), and a *ground segment* (OpenC3 and supporting services). In the simulation testbed cFS runs co-located with OpenC3 on the same Linux host, but logically belongs to the space segment. Attacks are organized into five defensive layers that drive the edge/central split:

- **L1 Network:** raw UDP/TCP attacks, source spoofing, port enumeration, passive interception (PoC-03, 12, 18, 20, 22).
- **L2 App Protocol:** Redis RESP abuse, HTTP/Traefik routing, MinIO S3 API, QuestDB SQL, CFDP PDU framing (PoC-01, 04, 06, 13, 17, 30).

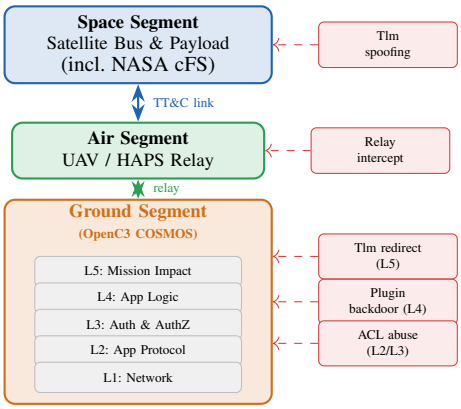


Fig. 1. SAGIN threat landscape and five defensive layers (L1–L5). NASA cFS is space-segment flight software; in the testbed it runs co-located with OpenC3 on the same Linux host. L1–L4 attacks are handled by edge SLMs; L5 attacks and multi-stage APT chains require the central LLM orchestrator.

- **L3 Auth & AuthZ:** Redis ACL, JWT tokens, service-account secrets (PoC-02, 05, 10, 14, 16).
- **L4 App Logic:** OpenC3 plugin/container manipulation, covert-tunnel routing, cFS integrity-check bypass (PoC-07, 08, 09, 11, 15, 23, 24, 29).
- **L5 Mission Impact:** telemetry falsification, schedule corruption, evidence suppression, flight-SW memory corruption, persistent flight-SW implants (PoC-19, 21, 25, 26, 27, 28).

L1–L4 attacks expose signals within a single software stack and are handled by the edge SLM co-located with that stack, locally and without depending on a live ground link. L5 attacks and multi-stage APT chains require cross-application or cross-segment reasoning that exceeds any single SLM's observability horizon, motivating the central LLM tier; this correlator works on compact ATT&CK-tagged summaries rather than raw telemetry, so the inter-tier channel fits within the contact window and bandwidth constraints documented in Section I.

PoC catalog and APT chains. Table I summarises the 30 PoCs by SAGIN segment, application, defensive layer, and primary detector tier. PoC-01–17 target the OpenC3 ground-station stack (Redis, Traefik, MinIO, QuestDB, plugin/script-runner); PoC-18–30 target the NASA cFS space-segment software (CI_LAB, TO_LAB, SCH_LAB, MM, ES, EVS, CS, CF/CFDP). The six deep-cFS exploits not previously reported in the literature are PoC-25 (EVS Stealth, T1562.001), PoC-26 (MM Arbitrary R/W, T1059), PoC-27 (CDS Implant, T1542.003), PoC-28 (ES Exec, T1059.004), PoC-29 (CS Bypass, T1562.001), and PoC-30 (CFDP Backdoor, T1105).

Five APT chains are defined by temporal sequencing of constituent PoCs within configurable windows: **R1 Recon-to-Hijack** (L3→L3→L5, PoC-05→02→19, cross-segment), **R2 Covert Persistence** (L4→L5→L5, PoC-09→27→25, cross-segment), **R3 Mission Blackout** (L5+L5+L5, PoC-19+21+25, cross-app within space segment), **R4 Recon-to-Destruction** (L2→L2→L2, PoC-17→01→13, cross-app within ground), and **R5 Plugin Backdoor** (L4→L4→L4, PoC-

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TABLE I
POC ATTACK CATALOGUE (OPENC3 + NASA CFS TESTBED). SEG: GND/SPC; L: DEFENSIVE LAYER; DET: PRIMARY DETECTOR TIER (SLM = EDGE; LLM = CENTRAL ORCHESTRATOR).

#	Seg	App	L	Det	Category	ATT&CK
Ground – OpenC3 (PoC-01–17)						
01	GND	Redis	L2	SLM	Config	T1562.001
02	GND	Redis	L3	SLM	ACL	T1078
03	GND	Net	L1	SLM	UDP Injection	T1190
04	GND	Redis	L2	SLM	Stream Poison	T1565.002
05	GND	Redis	L3	SLM	Recon Chain	T1552/87
06	GND	Traefik	L2	SLM	Hijack	T1574
07	GND	OpenC3	L4	SLM	Script Abuse	T1059.006
08	GND	OpenC3	L4	SLM	Plugin BD	T1195.002
09	GND	Docker	L4	SLM	Volume BD	T1543
10	GND	OpenC3	L3	SLM	Token Forge	T1528
11	GND	MinIO	L4	SLM	S3 Takeover	T1530
12	GND	Net	L1	SLM	TLS Sniff	T1040
13	GND	QuestDB	L2	SLM	DB Destruct	T1190
14	GND	OpenC3	L3	SLM	Token Abuse	T1565.001
15	GND	Docker	L4	SLM	Revive	T1543
16	GND	OpenC3	L3	SLM	Priv Esc	T1078
17	GND	OpenC3	L2	SLM	API Enum	T1595.001
Space – NASA cFS (PoC-18–30)						
18	SPC	CI_LAB	L1	SLM	CCSDS Replay	T1040
19	SPC	TO_LAB	L5	LLM	Tlm Redirect	T1565.002
20	SPC	CI_LAB	L1	SLM	Malformed Pkt	T1499.002
21	SPC	SCH_LAB	L5	LLM	Sch Hijack	T1053
22	SPC	CI_LAB	L1	SLM	Multi-CPU Scan	T1046
23	SPC	cFS bld	L4	SLM	Build Audit	T1068
24	SPC	OpenC3	L4	SLM	Route BD	T1572
25	SPC	EVS	L5	LLM	EVS Stealth	T1562.001
26	SPC	MM	L5	LLM	MM Arb R/W	T1059
27	SPC	CDS	L5	LLM	CDS Implant	T1542.003
28	SPC	ES	L5	LLM	ES Exec	T1059.004
29	SPC	CS	L4	SLM	CS Bypass	T1562.001
30	SPC	CF/CFDP	L2	SLM	CFDP BD	T1105

07→08→24, cross-state-dimension within a single application). All five exceed the observability horizon of any single edge SLM and motivate the central LLM tier.

IV. SKYGUARD ARCHITECTURE

Overview. SkyGuard organizes detection into two tiers (Figure 2). *Tier 1 (Edge):* per-stack SLM agents fine-tuned for agentic investigation of one software stack each. In the present deployment two agents are active: the OpenC3 GS agent (Redis/Traefik/MinIO/QuestDB/scriptrunner) and the cFS Proxy agent (CI/TO/SCH/MM/ES/EVS/CS/CF), both co-located on the simulation host `sat_gs`. The cFS Proxy is a simulation proxy because flight-qualified inference hardware is not yet available; the intended near-term deployment moves it to a HAPS-resident node intercepting TT&C relay traffic in-flight. *Tier 2 (Central):* a locally hosted Qwen2.5-32B-Instruct LLM correlates the structured, sanitized summaries from all edge agents, assembles APT chains, generates portable Tier B Sigma rules, and dispatches downward directives.

Tier 1: Edge SLM agents. Each SLM is a fine-tuned Qwen2.5-Instruct model (1.5B–7B parameters, 4-bit quantised) with access only to telemetry and logs generated within its own software-stack boundary. Qwen2.5 is selected across all tiers for its consistent native `<tool_call>` schema from 0.5B to 32B [30], allowing model size to serve as the

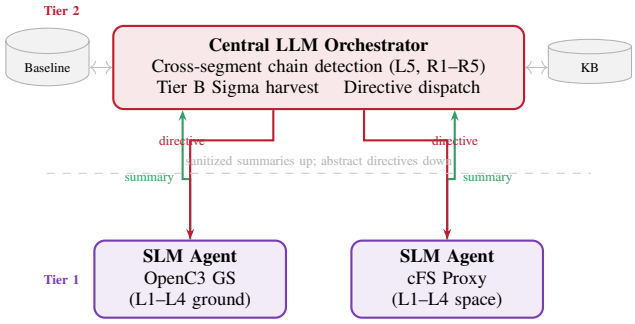


Fig. 2. SkyGuard two-tier architecture as deployed. The OpenC3 GS agent and the cFS Proxy agent are co-located on the simulation host `sat_gs`; the cFS Proxy stands in for a HAPS- or space-resident agent until flight-qualified inference hardware becomes available. The central LLM holds no concrete IOCs: only ATT&CK-tagged sanitized summaries flow up, and downward directives carry abstract behavioral patterns only.

sole independent variable in ablation. Each agent runs three complementary detection mechanisms tailored to SAGIN's bandwidth- and connectivity-constrained edge:

(i) Snapshot-based integrity verification. Periodically, or on receipt of a `HUNT` directive, the SLM captures current live state (ACL users, open ports, Docker volume mounts, cFS table files) and diffs it against a cryptographically signed known-good baseline. This tolerates contact-window gaps because only current state is required, and covers persistent state changes (PoC-02 new ACL user, PoC-09 Docker volume plant, PoC-27 CDS block modification).

(ii) Bounded-window log and event query. Several attack classes leave no diff: read-only reconnaissance (PoC-05, 17, 22) modifies nothing; transient attacks (PoC-20, 29, 30) complete between polls. For these the SLM queries finite, locally retained records on demand—Redis slow-log ring buffer, cFS EVS event log (a persistent write-ahead log of all cFS events), and a rolling UDP pcap. These are not live stream subscriptions; no data leaves the segment. The EVS log is the primary detection source for the entire PoC-18–30 cFS layer.

(iii) Semantic cross-dimension reasoning. Both mechanisms above can produce ambiguous signals. The SLM chains tool calls across snapshot diffs, slow-log queries, and EVS log reads until converging evidence across independent dimensions confirms or refutes a TTP hypothesis. PoC-02, for instance, is confirmed not by the ACL diff alone but by correlating the new account, its `+@all` privilege scope, and a slow-log `SETUSER` entry from a non-whitelisted source IP within seconds of a credential-harvest burst.

The output of each investigation is (i) an ATT&CK-tagged `CONFIRMED`/`SUSPECTED`/`NO_THREAT` verdict, (ii) a natural-language summary, (iii) a pre-redaction Tier A Sigma rule, and (iv) a hash reference to the locally retained forensic package.

Trajectory-level SFT. Because the SLM task is agentic investigation rather than static log classification, each agent is domain-adapted via *trajectory-level supervised fine-tuning*: every training example is a complete multi-turn tool-call trajec-

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tory in Qwen2.5's native format, comprising a system prompt, an initial trigger, the sequence of tool calls with their stated reasoning, the captured testbed tool outputs, and the final alert JSON. PoC replays are captured under instrumentation that records actual tool outputs (`redis-cli ACL LIST`, `stat`, `docker exec cfs grep`), so the SLM learns to interpret authentic system responses. Benign counterpart replays supply `NO_THREAT` trajectories. Fine-tuning uses LoRA (rank 16, α 32) on top of pre-trained Qwen2.5-Instruct via LlamaFactory [31].

Tier 2: Central LLM orchestrator. The central LLM is Qwen2.5-32B-Instruct (4-bit quantised, ≈20 GB VRAM) on a dedicated GPU server. Unlike the edge SLMs it is *not* fine-tuned: its inputs are already-investigated structured summaries rather than raw logs, making its task one of reasoning and synthesis where pre-trained capabilities outweigh narrow domain adaptation on a 30-PoC corpus (which would risk catastrophic forgetting of code generation and cross-domain reasoning). The central LLM is instead conditioned through prompt engineering plus retrieval-augmented generation (RAG) over ATT&CK for ICS [21] and SPARTA [22]. Few-shot Sigma examples are drawn from the 30 PoC ground-truth trajectories.

Two-tier Sigma harvest. SkyGuard bifurcates Sigma rule generation by redaction state:

- *Tier A (edge, pre-redaction).* Generated by the edge SLM from the full local forensic package before redaction. Contains real field values, log-source definitions, container names, and deployment-specific patterns; immediately deployable in the operator's SIEM. Retained locally; never transmitted.
- *Tier B (central, post-redaction).* Generated by the central LLM from the sanitized ATT&CK-tagged summary. Contains only technique-level detection logic and product-level log-source definitions (`product: redis`, `product: nasa-cfs`). Portable across SAGIN operators running the same software stack; suitable for ISAC/ATT&CK community sharing without revealing operational details.

Tier A is precise but not portable; Tier B is portable but requires the consuming operator to maintain an equivalent baseline artifact store. Both rules are harvested as a side-effect of every confirmed detection.

V. Tool-Calling and Hunt Confirmation

Information flow and confidentiality. Figure 3 summarises the data classification at each tier boundary. Raw operational data (**CONFIDENTIAL**) never leaves the edge segment: it is consumed only by the local SLM through its three detection mechanisms. Only structured ATT&CK-tagged summaries (**SANITIZED**) traverse the inter-tier channel upward (schema in Listing 1). Downward `HUNT`/`UPDATE_BASELINE` directives (**DIRECTIVE**) are signed (HMAC-SHA256) JSON objects with bounded TTL, verified by the SLM shim before any payload reaches the inference engine (schema in Listing 2).

Dual triggering modes. *Mode 1 (bottom-up, local).* The SLM polls local state and autonomously initiates an agentic investigation on a weak signal, runs its state-query tool-call chain to confirmation, generates a Tier A Sigma rule pre-redaction, and forwards the sanitized summary upward (Listing 3 for PoC-02 Redis ACL backdoor). The central LLM correlates the summary across segments, generates a Tier B portable Sigma rule, and may dispatch an `UPDATE_BASELINE` directive so that the newly confirmed malicious state is flagged on all future polls.

Mode 2 (top-down, directive). The central LLM has confirmed a partial chain from one segment and suspects another is involved in a stage not yet locally detected. It dispatches a `HUNT` directive carrying the chain context, ATT&CK technique IDs to investigate, and abstract behavioral `kb_patterns` drawn from the ATT&CK/SPARTA RAG corpus—but *no concrete IOCs*, because the central LLM holds only redacted summaries and cannot know the specific accounts, IPs, or paths in the target segment. The receiving SLM uses the semantic guidance as its starting hypothesis and discovers all concrete IOCs locally (Listing 4 for chain R1). This division of labor serves two purposes simultaneously: it keeps the upward channel small enough to fit within realistic contact-window throughput (Section I), and it ensures that the central LLM reasons at the ATT&CK technique level while the edge SLM operates at the forensic artifact level, so concrete IOCs and raw operational data never leave the segment they originate in.

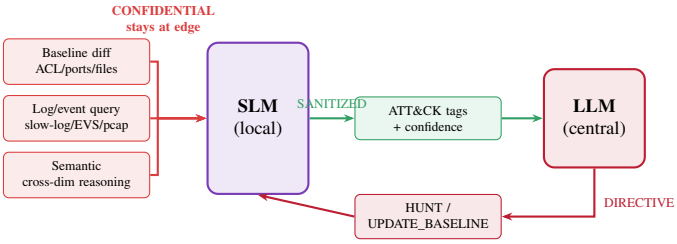


Fig. 3. SkyGuard information flow. Three local mechanisms (red) feed the edge SLM, which retains a Tier A Sigma rule locally and emits only ATT&CK-tagged summaries (green) upward. The central LLM generates a Tier B portable Sigma rule and dispatches HMAC-signed directives back to the edge.

Listing 1. Upward sanitized summary schema (edge SLM → central LLM). Raw IOCs are hashed or omitted before transmission; only ATT&CK-aligned semantic fields and integrity references cross the tier boundary.

```
{
  "segment":      "openc3_gs|cfs_proxy|...",
  "trigger":      "LOCAL|DIRECTIVE",
  "alert":        "CONFIRMED|SUSPECTED|NO_THREAT",
  "ttps":         ["T1136", "T1098", ...],
  "iocs":         ["<hashed account>", "<hashed IP>",
      ...],
  "severity":     "CRITICAL|HIGH|MEDIUM|LOW",
  "ts_utc":       "2026-05-04T03:17:42Z",
  "evidence_ref": "sha256:<local forensic package>",
  "chain_hint":   "<suspected APT chain ID, optional>"
}
```

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Listing 2. Downward directive schema (central LLM → edge SLM). HMAC-signed JSON with bounded TTL; `kb_patterns` carries abstract behavioral signatures from the ATT&CK/SPARTA RAG corpus, never concrete IOCs.

```
{
  "directive_id":   "<UUID for dedup/audit>",
  "directive_type": "HUNT|UPDATE_BASELINE",
  "target_segment": "openc3_gs|cfs_proxy|...",
  "chain_context":  "<confirmed stages at peer segment(s)
      >",
  "hunt_focus":     ["T1595", "T1046", ...],
  "kb_patterns":    ["<abstract behavioral pattern>",
      ...],
  "baseline_patch": "<signed diff, UPDATE_BASELINE only
      >",
  "ttl_seconds":    300,
  "hmac_sha256":    "<integrity signature>"
}
```

Listing 3. Mode 1 three-step SLM tool-call chain for PoC-02 Redis ACL backdoor (T1136/T1098). Each JSON block is emitted by the SLM; the indented comment shows the shim's execution result. Only after all three checks concur does the SLM emit `CONFIRMED`.

```
# Step 1 -- diff live ACL list against known-good
    baseline
{ "tool": "exec",
  "command": "docker exec openc3-redis redis-cli
    --user admin -a adminpassword ACL LIST
    > /tmp/current_acl.txt &&
    diff /opt/skyguard/baseline_acl.txt /tmp/current_acl.
        txt",
  "reason": "Detect accounts not in baseline
            {admin, openc3, scriptrunner, default}" }
  # result: +user svc_monitor on >password ~* +@all

# Step 2 -- inspect the unexpected account's permissions
{ "tool": "exec",
  "command": "docker exec openc3-redis redis-cli
    --user admin -a adminpassword
    ACL GETUSER svc_monitor",
  "reason": "Confirm +@all (full-access) privilege
      granted
            to backdoor account (T1098)" }
  # result: flags=on  passwords=1  commands=+@all  keys=*

# Step 3 -- check persistence timestamp on users.acl
{ "tool": "exec",
  "command": "docker exec openc3-redis
    stat /etc/redis/users.acl",
  "reason": "Establish write time of ACL SAVE to
            correlate with attack chain timeline (T1136)
                " }
  # result: Modify: 2026-05-04 03:17:42 UTC

# SLM final assessment (sanitized before transmit upward)
{ "alert":      "CONFIRMED",
  "segment":    "openc3_gs",
  "trigger":    "LOCAL",
  "ttps":       ["T1136", "T1098"],
  "iocs":       ["svc_monitor", "users.acl mtime 03:17:42
      "],
  "severity":   "CRITICAL",
  "chain_hint": "R1 -- correlate with ACL SETUSER
                 within 60s of recon burst (PoC-05)" }
```

VI. Experimental Evaluation

Testbed. All experiments run on a physical testbed with OpenC3 v5.x and NASA cFS v7.x in Docker Compose: a ground-station host (8-core, 32 GB) running Redis/Traefik/MinIO/QuestDB/scriptrunner, a cFS simulation host (4-core, 8 GB) running CI/TO/SCH/MM/ES/EVS/CS/CF, and a network emulator providing configurable latency and bandwidth. All 30 PoCs are scripted for automated replay. Edge SLMs are Qwen2.5-Instruct (1.5B, 3B, 7B; 4-bit quantised) fine-tuned via trajectory-level SFT with LoRA; results below use the 7B representative size. The central LLM is Qwen2.5-32B-Instruct (4-bit quantised, ≈20 GB VRAM).

Listing 4. Mode 2 `HUNT` directive for APT chain R1 (abbreviated). The central LLM holds only the redacted summary `ttps=[T1136,T1098]`, `chain_hint="R1 expect T1595/T1046"` from `openc3_gs`—no concrete IOCs. The receiving SLM discovers `172.20.0.14` and CMD_ERR_EID bursts locally.

```
# --- Central LLM dispatches HUNT directive ---
{ "directive_id":   "d7f3a1c2-...",
  "directive_type": "HUNT",
  "target_segment": "cfs_proxy",
  "chain_context":  "R1 partial: T1136/T1098
                     CONFIRMED openc3_gs",
  "hunt_focus":     ["T1595", "T1046"],
  "kb_patterns":    ["unexpected UDP burst to cFS
                      command ingest port (T1595)",
                     "CMD_ERR_EID rate spike (T1046)"],
  "ttl_seconds":    300, "hmac_sha256": "a3f9..." }

# --- SLM verifies HMAC + TTL, then investigates locally
    ---
# Step 1: count UDP packets to cFS command ingest port
tcpdump -r udp_recent.pcap 'udp and dst port 1234' \
  | wc -l    # -> 38; last src=172.20.0.14
# Step 2: check src against local baseline (unknown=
    anomalous)
grep '172.20.0.14' baseline_hosts.txt
  # -> (no match)
# Step 3: check cFS EVS for command error burst
docker exec cfs grep CMD_ERR_EID /cf/cfs_evs.log
  # -> 12 events at 03:17:44-48Z

# --- SLM final summary (sanitized before transmit) ---
{ "alert":"CONFIRMED", "segment":"cfs_proxy",
  "trigger":"DIRECTIVE",
  "ttps":["T1595","T1046"],
  "chain_hint":"R1 COMPLETE -- T1136/T1098+T1595/T1046" }
```

Baselines. Four baselines form a 2×2 ablation over the presence of the edge and central tiers:

- **B1 – Static Snapshot Scripts:** human-authored shell scripts polling system state against hardcoded known-bad conditions. Isolates the value of LLM reasoning over baseline diffing.
- **B2 – Edge SLMs only:** edge SLMs perform agentic investigation identically to B4, but no central tier and no chain correlation. Isolates the value of the central LLM tier.
- **B3 – Central LLM only (infeasible reference):** a single central LLM with unredacted remote tool execution on all segments. This baseline assumes a persistent, high-bandwidth, low-latency link between the central analyzer and every segment—a premise that, as argued in Section I, the SAGIN link budget does not satisfy [5], [7], [8]. We include B3 not as a deployable alternative but as a ceiling that quantifies how much detection signal would be available *if* the link constraints did not exist; the gap between B4 and B3 measures the cost of respecting them.
- **B4 – SkyGuard (proposed):** full two-tier architecture with redaction, chain correlation, and dual-mode directives, operating within the realistic SAGIN link budget.

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Protocol. Results use leave-one-out cross-validation (LOO-CV) over the 30-PoC catalog: for each PoC the SLM is fine-tuned on the remaining 29 trajectories and evaluated on the held-out PoC; the held-out trajectory is also removed from the central LLM's RAG corpus. Recall is averaged over the 30 folds (per-PoC) or the three folds in which a chain constituent is the test PoC (per-chain). B1 uses neither fine-tuning nor the RAG corpus. Each (PoC, baseline) pair runs one attack and one benign counterpart from a clean snapshot. Under strict scoring only `CONFIRMED` with a matching ATT&CK TTP counts as TP. All baselines achieve P = 1.00 (FP = 0) across all categories, so we report recall and F1.

Detection effectiveness. Table II reports per-category recall and F1 under LOO-CV; Table III reports the analogous breakdown for the five APT chains. Three findings stand out.

Static rules are blind to read-only, transient, and in-memory attacks. B1 collapses to zero recall on L1 Network and L3 Auth & AuthZ because PoC-05/17/22 leave no persistent state diff (read-only reconnaissance), PoC-18/20 complete between polls (transient), and PoC-26 corrupts only RAM (no file-system artifact). B2 recovers most of these by chaining bounded-window log queries (mechanism ii) and semantic reasoning (mechanism iii)—reading the Redis slow-log, EVS event log, and rolling pcap at investigation time. The macro-recall gap (B1 = 0.30 vs. B2 = 0.46) is entirely a recall problem arising from the three blind spots above; the in-coverage per-category gap is largest on L1–L3 (no diff-able artifact) and smallest on L4 OpenC3 (persistent file artifacts).

Application coverage in training data is the dominant driver of SLM recall. B2 and B4 share the same edge SLMs and therefore the same per-category structure: high recall when the held-out PoC's application has at least one same-layer sibling in the training fold (L1 CI_LAB cluster, L3 Redis cluster, L4 OpenC3 cluster), and zero recall when the application is unique in the catalog (L4 cFS, L5 Mission Impact, and the L2 orphans Traefik/QuestDB/CF-CFDP). B3 retains the highest macro recall (0.82) because its 32B parametric knowledge of public applications partially substitutes for the missing RAG trajectory.

The 0.00 rows are a LOO/coverage decoupling artifact, not a detection-mechanism failure. In the L4-cFS and L5-Mission Impact rows of Table II, and the R3 row of Table III, every PoC targets an application that appears in exactly one PoC across the entire 30-PoC catalog (cFS-build, CS, TO_LAB, SCH_LAB, EVS, MM, CDS, ES). LOO-CV deliberately removes that one occurrence from training, leaving zero same-application examples for the edge SLM and therefore zero recall by construction. The tool-call chains that detect these PoCs in-distribution are documented per-PoC in Section V (e.g., PoC-25 EVS Stealth queries the EVS filter configuration directly to detect the *absence* of expected events; PoC-27 CDS Implant adds a restart step and post-restart block diff; PoC-23 Build Audit uses a third triggering mode for periodic compliance polling), and the B3 column confirms recall in the 0.63–0.65 band when the constraint is relaxed.

TABLE II
PER-CATEGORY DETECTION UNDER LOO-CV (7B EDGE SLM). P = 1.00 FOR ALL BASELINES; R AND F1 REPORTED. OVERALL ROW IS MACRO-AVERAGED. ROWS MARKED † ARE ORPHAN-APPLICATION CATEGORIES (EVERY POC TARGETS AN APPLICATION APPEARING NOWHERE ELSE IN THE CATALOG), SO LOO-CV LEAVES ZERO SAME-APPLICATION TRAINING EXAMPLES AND B2/B4 COLLAPSE TO 0.00 BY CONSTRUCTION; SEE PROSE DISCUSSION.

		B1		B2		B3		B4	
Category	N	R	F1	R	F1	R	F1	R	F1
L1 Net/UDP	5	.00	.00	.80	.89	.95	.97	.80	.89
L2 App Protocol	6	.17	.29	.30	.46	.80	.89	.30	.46
L3 Auth & AuthZ	5	.00	.00	.90	.95	.95	.97	.90	.95
L4 OpenC3	6	.83	.91	.75	.86	.90	.95	.75	.86
L4 cFS†	2	.50	.67	.00	.00	.65	.79	.00	.00
L5 Mission Impact†	6	.33	.50	.00	.00	.64	.78	.00	.00
Overall macro	30	.30	.40	.46	.53	.82	.89	.46	.53

TABLE III
PER-CHAIN DETECTION UNDER LOO-CV (7B EDGE SLM). P = 1.00 FOR ALL BASELINES. B1/B2 SCORE ZERO ON EVERY CHAIN (NO TEMPORAL CORRELATION / NO CENTRAL TIER); B3 IS AN INFEASIBLE REFERENCE (ASSUMES UNREDACTED PERSISTENT HIGH-BANDWIDTH LINK TO EVERY SEGMENT). R3† IS AN ALL-ORPHAN-APPLICATION CHAIN AND COLLAPSES BY THE SAME CONSTRUCTION AS THE L5 ROW OF TABLE II.

		B1		B2		B3		B4	
Chain	N	R	F1	R	F1	R	F1	R	F1
R1 Recon-to-Hijack	3	.00	.00	.00	.00	.87	.93	.60	.75
R2 Covert Persist	3	.00	.00	.00	.00	.70	.82	.30	.46
R3 Mission Blackout†	3	.00	.00	.00	.00	.63	.77	.00	.00
R4 Recon-to-Destruct	3	.00	.00	.00	.00	.85	.92	.33	.50
R5 Plugin Backdoor	3	.00	.00	.00	.00	.95	.97	.90	.95
Overall macro	15	.00	.00	.00	.00	.80	.88	.43	.53

We retain the 0.00 numbers as an honest out-of-distribution stress test of training-coverage; the explicit fix—a closed-loop pipeline that augments the corpus with same-application sibling trajectories—is reported as the lead future-work item below.

Chain detection requires both tiers, and SkyGuard is the only deployable configuration that achieves it. B1 and B2 both score R = 0.00 on every chain: B1 has no cross-event correlation, and B2 emits per-PoC summaries but no central tier exists to assemble them. B3 achieves 0.80 macro chain-recall but only by assuming an unredacted, persistent, high-bandwidth link to every segment—a premise the SAGIN link budget does not satisfy [7], [8]. Among the three deployable configurations (B1, B2, B4), SkyGuard is the only one that detects multi-stage chains at all: B4 reaches 0.43 macro chain-recall under LOO-CV, decomposing as 0.90 on R5 (where all three constituents target OpenC3 L4 and back each other up), 0.60 on R1, 0.33 on R4, 0.30 on R2, and 0.00 on R3. The R3 collapse reflects an edge-SLM application-coverage gap discussed in the next paragraph rather than a limit of the hierarchy itself; even so, the gain from B2 to B4 (0.00 $\rightarrow$ 0.43 macro chain-recall) isolates the value of the central-tier correlator under the link constraints.

Worked example: R1 Recon-to-Hijack. Tracing R1 (PoC-

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05→02→19, L3→L3→L5, 5-min window) through the four baselines makes the contrast concrete. At $T{+}0$ PoC-05 issues `ACL LIST`, `CONFIG GET *`, `SCAN *token*`—read-only credential harvest. At $T{+}90$s PoC-02 creates `svc_monitor` with `+@all`. At $T{+}4$min PoC-19 issues `TO_LAB ENABLE_OUTPUT` to redirect mission telemetry to `10.0.0.99`. *B1* detects only two isolated signals (new ACL user; EVS log entry) but cannot reach `CONFIRMED` on either because `svc_monitor` resembles a legitimate account and the `TO_LAB` change could be operator-authorised. Score: 0 TP, 0/1 chain. *B2* confirms all three PoCs locally—the Redis SLM correlates the slow-log burst with the `ACL SETUSER` entry from the same source IP, and the cFS SLM cross-checks the `TO_LAB` destination against the whitelist—but with no central tier the cross-stack link is never made. Score: 3 TP, 0/1 chain. *B3* sees everything and assembles R1 directly—if the SAGIN link could sustain raw-telemetry streaming from every segment to a central analyzer, which it cannot [8]. Score: 3 TP, 1/1 chain (infeasible reference). *B4* confirms all three constituents at the edges, both SLMs embed `chain_hint=R1`, and the central LLM assembles R1 from the TTP sequence and timestamp proximity alone—without ever seeing `172.20.0.14`, the harvested credentials, or the `TO_LAB` command payload, and without requiring more than a few hundred bytes of upward traffic per investigation. Score: 3 TP, 1/1 chain. On this Redis fold B4 reproduces the infeasible-reference result while operating within the SAGIN link budget; the cost of operating within that budget surfaces only on chains with application-unique constituents at one or more positions (TO_LAB-fold, CDS-fold, EVS-fold), where edge-SLM training coverage rather than the architecture itself becomes the limiting factor.

Latency and link budget. End-to-end detect-to-alert latency for an APT chain in B4 is bounded by the slowest edge SLM plus the central LLM correlation step—typically 4–6 s with a 7B edge SLM and the 32B central LLM [32], well within the seconds-to-minutes chain windows for R1–R5. On the link budget: the upward summary (Listing 1) compresses to $S \approx 500\,\mathrm{B}$; a $T = 5\,\mathrm{min}$ LEO pass at $R = 10\,\mathrm{kb/s}$ admits $RT/S \approx 750$ summaries, versus the ≈ 10 per pass produced by two edge agents at one investigation per agent per minute—a headroom of $\sim 10^3$. Directives (Listing 2) are comparable. A B3-style analyzer would instead have to egress the raw slow-logs, EVS streams, and pcaps that those summaries condense, exceeding the same window's capacity by orders of magnitude on any published LEO budget [5]–[7]. The hierarchy is thus quantitatively necessary, not only architecturally motivated.

VII. Discussion

Scope of the claims. Section I distinguishes the architectural-necessity argument (a reframing of an existing pattern for the SAGIN link budget) from the two design mechanisms and the artifact contribution that are, to our knowledge, new (the locality-preserving directive regime, the two-tier Sigma harvest, and the reproducible 30-PoC OpenC3+cFS catalog with six new deep-cFS exploits). We do not claim novelty for hierarchical LLM organization, multi-agent LLM coordination, RAG, tool-calling agents, or single-tier LLM-based Sigma-rule generation; each exists in the primary sources we cite, and the trajectory-level SFT recipe is a methodological adaptation rather than a new technique.

Limitations. Trajectory-level SFT requires expert-verified tool-call sequences and is labor-intensive; the 30-PoC corpus yields ≈600 augmented trajectories after IP, timing, and payload variation, but remains small by LLM fine-tuning standards. The cFS Proxy is a simulation proxy on the same Linux host as the OpenC3 GS agent: radiation-hardened flight computers (typically 300–800 MHz, 2–8 GB ECC RAM, 5–30 W) cannot sustain SLM inference today. The intended near-term path is a HAPS-resident SLM that intercepts TT&C relay traffic, recovering the logical space-segment placement at the cost of exposing raw telemetry to the relay node before redaction. Tier B Sigma rules depend on the consuming operator maintaining an equivalent baseline artifact store—an intentional incentive for baseline-first hygiene but a barrier to immediate deployment. A sophisticated attacker aware of SkyGuard's detection logic could craft state changes outside the SLM's training distribution or corrupt the baseline itself; dedicated adversarial robustness evaluation is left to future work.

Future work. The lead extension, motivated directly by the 0.00 rows in Tables II and III, is a *closed-loop adversarial fine-tuning pipeline* that converts every confirmed detection into additional training signal. An LLM coding assistant synthesizes novel variants of each PoC under expert supervision, the testbed replays them to harvest trajectories in the same schema used in this work, and the resulting examples are added to the edge-SLM fine-tuning corpus. This is the explicit remedy for the orphan-application problem identified above: it populates the currently single-PoC applications (cFS-build, CS, TO_LAB, SCH_LAB, EVS, MM, CDS, ES) with same-application sibling trajectories so that LOO-CV no longer leaves the held-out fold empty. Based on the within-application generalization profile observed for non-orphan categories in Table II (one same-layer sibling moves recall into the 0.6–0.9 band) and the B3 ceiling already reached for these categories (0.64–0.65), we expect the augmented corpus to lift L4 cFS and L5 Mission Impact recall toward the B3 envelope and to bring R3 chain recall in line with the other chains. Other planned work includes reinforcement learning from hunt-execution feedback [24], federated fine-tuning of edge SLMs across operators without raw-telemetry sharing, evaluation of a HAPS-resident agent to quantify the locality–functionality trade-off characterized in the Limitations, and cross-family SLM ablation (Phi-3.5-mini, Mistral-7B) beyond the Qwen2.5 size sweep used here.

VIII. Conclusion

We presented SkyGuard, the first realization of a capacity-graded SLM/LLM threat hunting hierarchy, specialized to the OpenC3+NASA cFS open-source space-software stack. Two co-located edge SLM agents (OpenC3 GS and cFS Proxy)

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cover L1–L4 detection locally within each software stack's observability boundary; a central LLM orchestrator handles L5 mission-impact assessment and APT-chain correlation across segments. The hierarchy is architecturally necessary rather than stylistically preferred: short LEO contact windows, bandwidth-rationed uplinks, and the established infeasibility of ground-based real-time IDS for intra-satellite threats [8] all rule out the central-analyzer pattern. A locality-preserving directive design ensures the central LLM never holds concrete IOCs and that upward traffic fits within realistic contact-window throughput; abstract behavioral patterns from ATT&CK and SPARTA guide receiving SLMs without exposing cross-segment operational details. Validation on 30 PoC attacks—including six previously unreported deep-cFS exploits and five APT chains—under leave-one-out cross-validation demonstrates that SkyGuard is the only deployable configuration evaluated that simultaneously detects multi-stage APT chains and operates within the SAGIN link budget; a static-scripts baseline and an edge-SLM-only baseline both score zero on every chain, while an infeasible full-information central-LLM reference achieves higher absolute recall under an architectural premise that SAGIN cannot satisfy. The remaining gap relative to that infeasible reference is dominated by application coverage in the edge SLM training distribution and motivates the closed-loop adversarial fine-tuning extension planned as immediate future work.

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